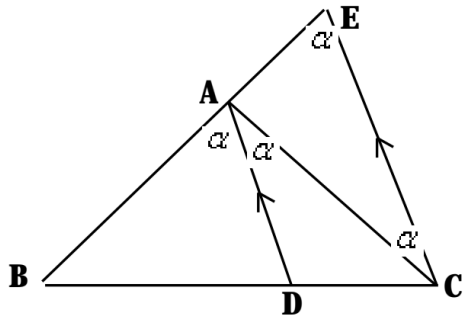
9. Bisector Theorem

Theorem

(a) **The internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle bisected.**

(b) **The external bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle bisected.**

(a)



(a)

Given : **AD is the internal bisector of the angle BAC of the triangle ABC and meets BC at D.**

To prove : $\frac{AB}{AC} = \frac{BD}{CD}$

Construction : **From C draw CE parallel to DA to meet BA produced at E.**

Proof : $AD \parallel CE$

$\angle DAC = \angle ACE$ (Alternate angles)

$\angle BAD = \angle AEC$ (Corresponding angles)

$\angle DAC = \angle BAD$ (Given)

Therefore $\angle ACE = \angle AEC$

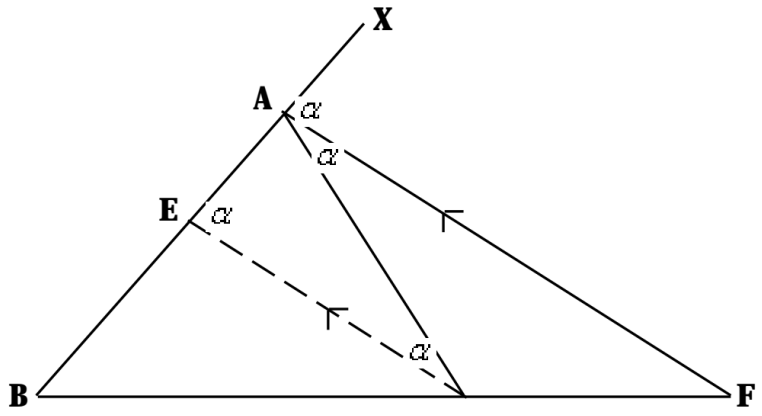
$AE = AC$

Since AD is parallel to EC

$$\frac{BA}{AE} = \frac{BD}{DC} \quad \text{and} \quad AE = AC$$

$$\frac{AB}{AC} = \frac{BD}{CD}$$

(b)



Given : **AF is the external bisector and meets BC produced, at F.**

To prove : $\frac{AB}{AC} = \frac{BF}{CF}$

Construction : **From C draw CE parallel to FA to meet BA at E**

Proof : $CE \parallel FA$

$\angle ECA = \angle CAF$ (Alternate angles)

$\angle CEA = \angle FAX$ (Corresponding angles)

But $\angle CAF = \angle FAX$

Therefore $\angle ECA = \angle CEA$

$AE = AC$

In triangle BAF, $CE \parallel FA$

$\frac{BF}{CF} = \frac{BA}{AE}$ and since $AE = AC$

$\frac{BF}{CF} = \frac{BA}{AC}$

Exercise 9

1 In a quadrilateral PQRS, PQ // SR. PR and QS intersect at T. Prove that

- $\Delta PQT \parallel \Delta SRT$
- $\frac{PR}{PT} = \frac{QS}{QT}$

2 In a triangle ABC, the interior and the exterior bisectors of the angle meet BC at X and Y respectively. If AB = 7.2 cm, AC = 5.4 cm, BC = 3.5 cm

- Prove that $BX : XC = 4 : 3$
- Find the ratio $BY : YC$

3 PS is a median of triangle PQR. Bisectors of the angle PSQ, and PSR meet PQ and PR at L and M respectively. Prove that LM // QR.

4 In a quadrilateral ABCD, the bisectors of the angles BAC and DAC meet BC and CD at L and M respectively. Prove that LM // BD.

5 If I is the in-centre of the triangle PQR, and if PI is produced to meet BC at X, show that

$$PI : IX = PQ + PR : QR$$

6 AD is a median of a triangle ABC. E is a point on produced AD. The bisector of $B\hat{D}E$ meets produced AB at H and the bisector of $C\hat{D}E$ meets produced AC at K. Show that HK // BC.